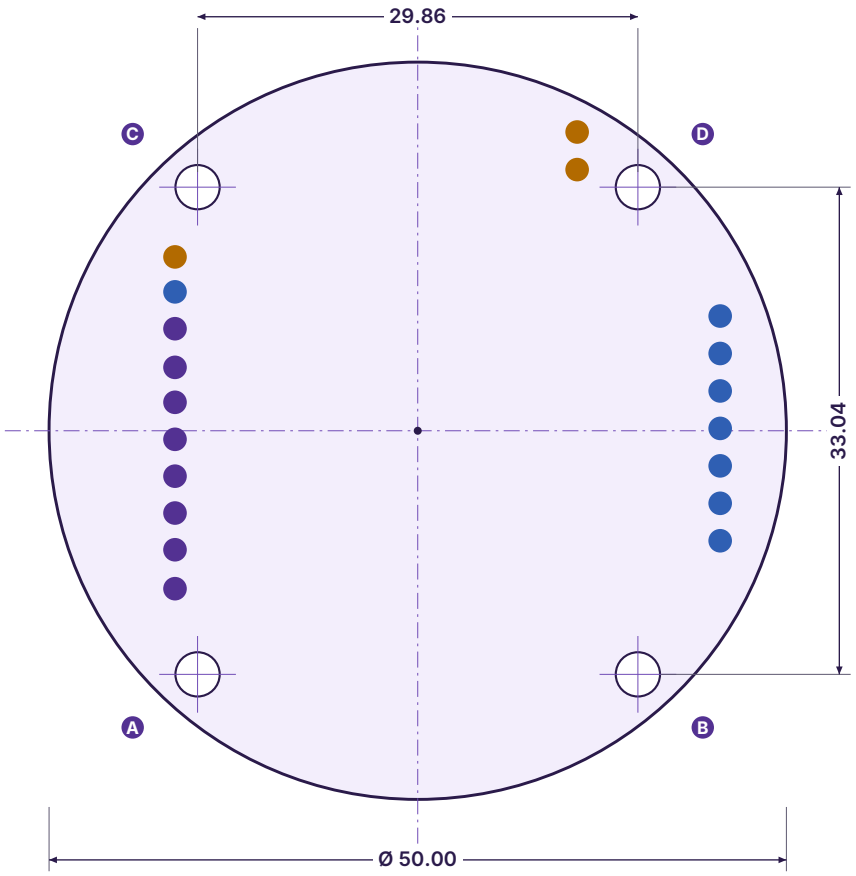
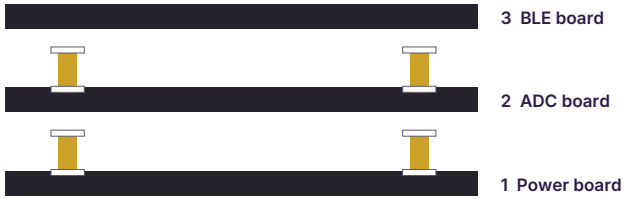
TOP VIEW · ADC-BOARD ORIENTATION (STACK REFERENCE)



- Electrode header E1–E8 (1 × 10)
- BIAS pin
- REF pin / 2-pin power header
- 7-pin board-to-board header
- 4 × Ø3.0 holes, 29.86 × 33.04



Exploded side view (schematic). M2 × 3 mm brass spacers with a washer at every board face.
Battery connector on the underside of the power board.

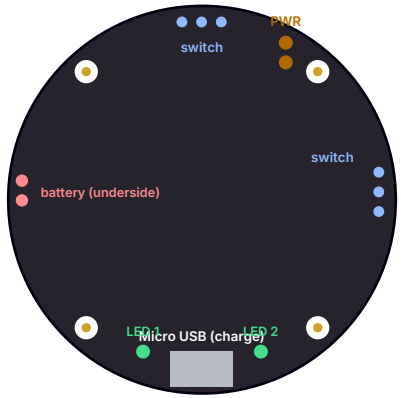
Title	IronBCI-8 stack outline & interfaces
Doc. No.	IB8-MD-002
Revision	Rev. A · 10 Nov 2026
Boards	Power V2.6 · ADC V3.2 · BLE MK3.0
Units	mm; datum = board centre
Scale	1.95 : 1 (A4 print)

Notes

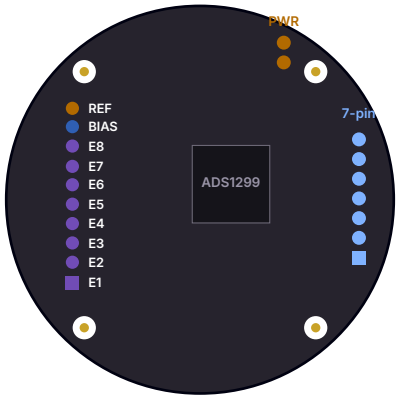
- All three boards share the Ø 50.0 mm outline and the same 4-hole pattern; holes A–D at (±14.93, ±16.52) from the centre.
- Board thickness 1.6 mm nominal per board (confirm with the fabricator data before ordering a case).
- Electrode header: 10 pins, 2.54 mm, at X −16.46; E1 nearest hole A, Ref nearest hole C.

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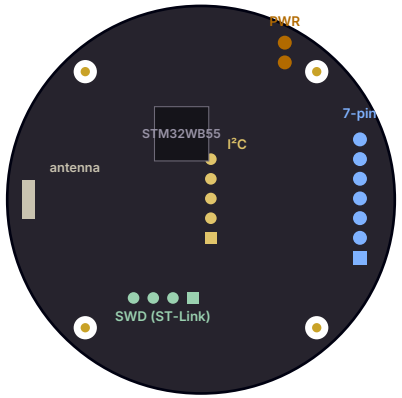
02 BOARD VIEWS AND INTERFACE LOCATIONS



Power board (bottom)



ADC board (middle)



BLE board (top)

Figure 2 — The three boards seen from above, each rotated to its position in the stack. Pin 1 shown square.

Board	Interface	Position (stack view, mm)	Details
ADC	**Electrode header**	X −16.46, Y −10.71 (E1) … +11.79 (Ref)	1 × 10, 2.54 mm: E1–E8, Biasout, Ref
ADC / BLE	**Board-to-board header**	X +20.51, Y −7.46 … +7.78	7 pins, internal link — not for user connection
All	**Power header**	X +10.81, Y +17.71 / +20.26	2 pins (GND, 5 V), passes through the stack
BLE	**SWD header J2.1**	bottom edge, near hole A–B	4 pins: RST, SWDIO, GND, SWCLK (ST-Link)
BLE	**I²C header J2.4**	centre, X ≈ +1.3	5 pins: Reserved, GND, 3.2 V, SCL, SDA
Power	**Micro USB**	bottom edge, centre	Charging only, 5 V; LED 1 / LED 2 either side
Power	**Switches 1 and 2**	top edge and right edge	Slide switches: battery and data flow
Power	**Battery connector**	left edge, underside	2.0 mm JST, 2 pins; check GND marking

Hole	X	Y	Ø	Use
A	-14.93	-16.52	3.00	M2 + washer
B	+14.93	-16.52	3.00	M2 + washer
C	-14.93	+16.52	3.00	M2 + washer
D	+14.93	+16.52	3.00	M2 + washer

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03 STACK ASSEMBLY

1 Prepare the hardware

4 × M2 × 3 mm spacers per board joint, M2 screws (2–3 mm) and plastic washers. Work on an anti-static mat.

2 Power board at the bottom

Micro USB and switches face outwards; battery connector on the underside. Fit the four spacers with a washer under each.

3 ADC board in the middle

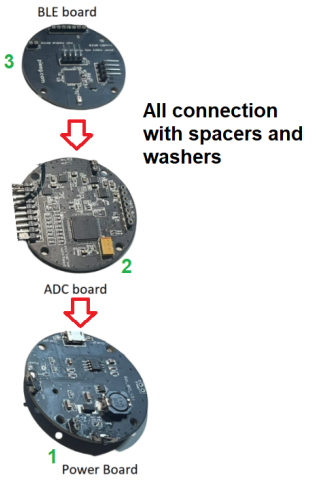
Align the 2-pin power header and press it down; electrode header to the left as in Figure 2. Washer on both faces of the board.

4 BLE board on top

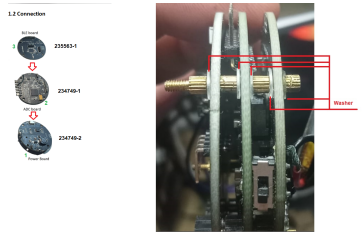
Align the 7-pin and 2-pin headers with the ADC board and press evenly until fully seated. Fit the top screws with washers.

5 Check

Every header pin fully engaged, no board tilted, nothing touching the antenna at the left edge of the BLE board.



Stack order: BLE (3), ADC (2), power (1).



Spacers and washers between boards.

CAUTION Keep the antenna clear.

No metal screws, cables or conductive parts within a few millimetres of the chip antenna. Use plastic washers; a case should be plastic (no carbon or metal fill).

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